

⚡ ENTERPRISE ENGINEERING MANUAL

XortLogix High Level Assistant & AI Platform

Comprehensive Technical Architecture, Hybrid RAG Retrieval Engine, GoHighLevel REST API v2 & OAuth 2.0 Integration, and Custom Front-End Deliverables.



Hybrid RAG Vector Store

5,717 semantically indexed chunks of official documentation, FastEmbed ONNX Nomic v1.5, and Reciprocal Rank Fusion.



Multimodal Intelligence

Native document parsing (PyPDF), clipboard screenshot vision analysis (Ctrl+V), and real-time audio voice notes.



Custom Frontend Deliverables

SPA DOM lifecycle management, debounced MutationObserver, routeChangeEvent handling, and dark theme injection.



Zero-Hallucination Guardrails

System-level strict proposal rules, source separation protocol, and automatic [CANDIDATE INPUT REQUIRED] flagging.

Developed For: XortLogix Technical Solutions Group

Backend Architecture: FastAPI 0.115 • ChromaDB • Google Gemini
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Target Audience: This document is prepared for software engineers, technical solution architects, product managers, DevOps teams, and clients seeking an exhaustive, transparent view of the XortLogix High Level platform architecture, data pipelines, and customization capabilities.

1 Executive Summary & System Purpose

The **XortLogix High Level Assistant** is an enterprise-grade artificial intelligence platform engineered to bridge the technical divide between complex GoHighLevel CRM infrastructure, custom software development, and senior-level technical solutions architecture.

1.1 The Challenge in GoHighLevel Engineering

GoHighLevel (HighLevel / LeadConnector) has become the dominant all-in-one CRM and marketing automation engine for modern agencies and SaaS enterprises. However, delivering advanced technical solutions on HighLevel introduces severe technical barriers:

- **API Evolution & Deprecation:** HighLevel officially deprecated API v1 on December 31, 2025. Standard generic LLMs (such as base ChatGPT) frequently generate broken v1 snippets with legacy endpoints and incorrect header formats.
- **Strict OAuth 2.0 & Token Lifecycles:** Multi-tenant HighLevel apps require strict 24-hour token refresh rotation, granular permission scopes, and specific HTTP request headers ([Version: 2021-07-28](#)).
- **Dynamic SPA Limitations:** HighLevel renders dashboards, contacts, and funnels as a client-side Single Page Application (SPA). Standard static scripts fail because target DOM nodes are rendered dynamically as users navigate.
- **AI Hallucination in Consulting Proposals:** When generating technical RFPs or proposals, standard AI models frequently fabricate experience, claim non-existent case studies, or invent metrics, undermining credibility with enterprise clients.

1.2 The Platform Solution

XortLogix High Level Assistant solves these hurdles by providing a unified, multimodal technical co-pilot backed by a factual vector retrieval engine:

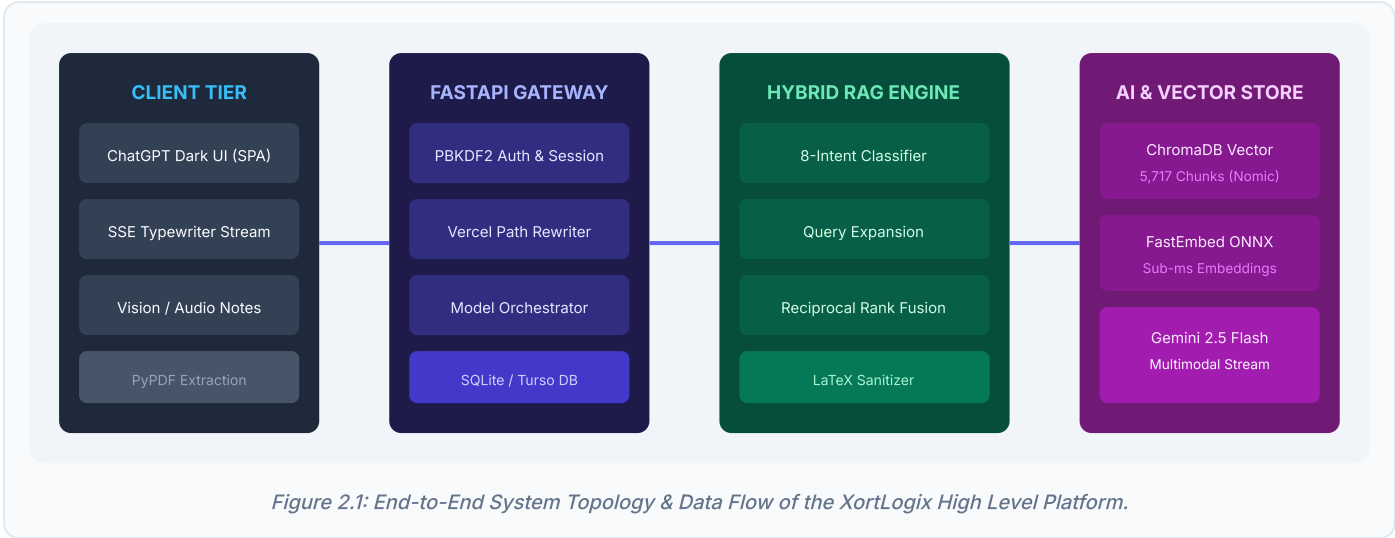
Key Innovation: Instead of relying solely on general LLM weights, the assistant interrogates a curated 5,717-chunk ChromaDB vector store representing official REST API v2 docs, OAuth handshakes, and proprietary company frontend code before synthesizing responses.

1.3 Core Business & Technical Objectives

Functional Domain	Native GHL Limitation	XortLogix Platform Deliverable
Technical Support	Overwhelmed support agents; slow API documentation search	Instant RAG answers with exact endpoints, parameters, and payloads
Custom Frontend	No native UI customization or dynamic custom widgets	Battle-tested JS (MutationObserver / SPA route listeners) & CSS themes
RFP & Proposal Generation	Generic, inaccurate, and hallucinated AI bids	Strict proposal rules separating verified facts from recommendations
Multimodal Diagnostics	Cannot interpret screenshots or audio walk-throughs	Live audio notes, PDF extraction, and clipboard screenshot vision analysis

2 High-Level System Architecture & Tech Stack

The system is organized into decoupled layers to guarantee low latency, horizontal scalability, and zero downtime across both serverful and serverless cloud environments.



2.1 Component Technical Specifications

Module	Source File	Primary Responsibility	Key Dependencies
Core Server	<code>app.py</code>	FastAPI app, SSE streaming, authentication cookies, static asset serving, warmup thread.	<code>fastapi</code> , <code>uvicorn</code> , <code>pydantic</code>
RAG Engine	<code>rag_engine.py</code>	Query analysis, intent classification, hybrid retrieval, RRF scoring, prompt scaffolding.	<code>fastembed</code> , <code>chromadb</code>
Database Engine	<code>db.py</code>	User accounts, PBKDF2 hashing, sessions, conversations, Turso cloud HTTP client & SQLite WAL.	<code>sqlite3</code> , <code>urllib.request</code>
RAG Microservice	<code>rag_microservice.py</code>	Lightweight secondary API on port 7861 for external PHP Laravel or Node.js web frameworks.	<code>fastapi</code> , <code>google-genai</code>
Frontend Engine	<code>static/app.js</code>	SPA UI controller, SSE reader, media recorder, lightbox preview, history date-grouping.	Vanilla JS (ES6+)

3 Hybrid RAG Engine & Vector Intelligence Deep Dive

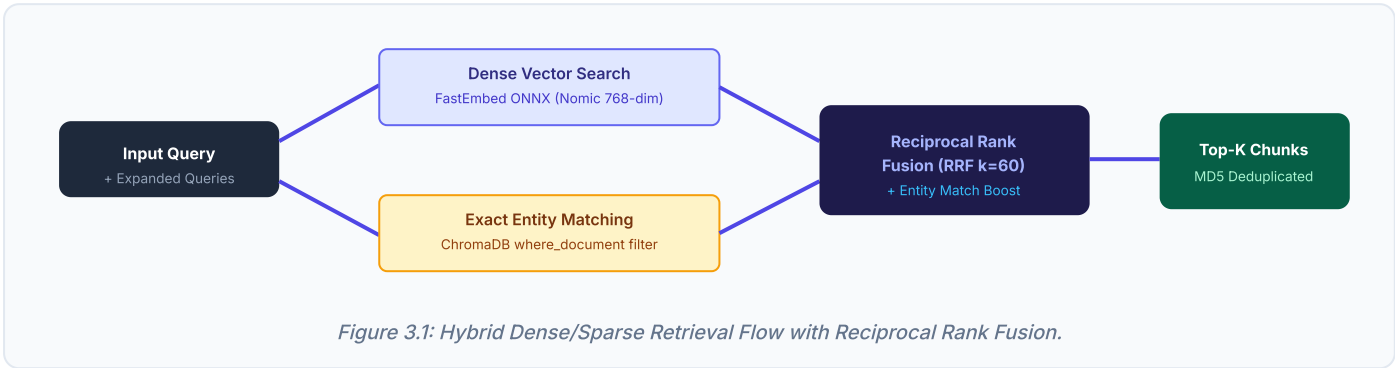
The core intelligence of the platform resides in `rag_engine.py`. It is engineered to overcome the classic limitations of naive semantic search by integrating query understanding, multi-query expansion, hybrid dense/sparse retrieval, and Reciprocal Rank Fusion.

3.1 Query Understanding & The 8 Intent Classes

Upon receiving a user query, the engine executes `QueryUnderstandingEngine.analyze()` before invoking vector search. The query is classified into one of eight distinct intents:

Intent Class	Trigger Criteria / Keywords	System Action & Output Structure
<code>conversational_greeting</code>	"hi", "hello", "salam", "aoa", "good morning", "who are you"	Bypasses vector search completely. Streams zero-latency personalized greeting with user's first name.
<code>out_of_scope</code>	Keywords: recipes, weather, gaming, movies, sports, trivia	Bypasses LLM generation. Outputs standardized Out-of-Scope notice declaring exclusive GHL focus.
<code>frontend_customization</code>	"MutationObserver", "routeChangeEvent", "formula404", "custom JS", "dashboard theme"	Applies multi-query expansion. Generates 100% complete IIFE JavaScript blueprint with DOM lifecycle handlers.
<code>job_posting_analysis</code>	RFP text, "we are looking for", "responsibilities", "compensation", "KPIs"	Analyzes technical needs, missing gaps, and generates structured proposal strategy adhering to strict factual rules.
<code>technical_troubleshooting</code>	"error", "failed", "bug", "404", "500", "webhook not firing"	Identifies root cause, probable failure points, and outputs step-by-step resolution in GoHighLevel.
<code>system_architecture</code>	"how to build", "how to setup", "booking funnel", "workflow", "pipeline"	Generates complete system blueprint with Triggers, Actions, Custom Values, and Edge-Case safeguards.
<code>comparison_evaluation</code>	"difference between", "vs", "versus", "compare", "which is better"	Compares features, limitations, cost implications, and provides concrete recommendations.
<code>factual_lookup</code>	Standard technical questions, endpoint lookups, header specifications	Returns concise, direct factual explanation based on official documentation context.

3.2 Hybrid Dense & Sparse Retrieval Architecture



3.3 The Composite Ranking Equation & Anti-Hallucination Guardrails

```
Composite_Score = (RRF_Score * 2.0) + (Vector_Score * 0.5) + (Keyword_Score * 0.3) + (Entity_Matches * 0.1)

Where:
- RRF_Score = Sum( 1.0 / (60 + rank + 1) ) across original and expanded queries
- Vector_Score = 1.0 / (1.0 + Cosine_Distance)
- Keyword_Score = 0.3 for each exact entity match in chunk text
- Entity_Matches = Count of core query entities identified in chunk content
```

1. **Zero Experience Fabrication:** Never invents previous clients, metrics ("scaled to \$500K/mo"), or portal counts ("35+ portals").
2. **Strict Source Separation:** Always categorizes outputs into (A) Verified Experience, (B) Client Requirements, and (C) Proposed Solutions.
3. **Mandatory Flagging of Missing Data:** Flags unverified facts strictly as `[CANDIDATE INPUT REQUIRED]`.
4. **LaTeX Sanitization Engine:** `clean_latex_artifacts()` replaces all raw math delimiters and `\rightarrow` with unicode arrows (`→`).

4 Knowledge Ingestion & Web Scraping Pipeline

The factual foundation of the XortLogix platform is its custom-indexed knowledge base, containing 5,717 dense vector chunks covering the complete GoHighLevel developer surface.

4.1 Multi-Threaded Sitemap Scraper (`scrape\_ghl\_docs.py`)

```
# scrape_ghl_docs.py - High-Performance Concurrent Scraping Pipeline
SITE_MAP_URL = "https://marketplace.gohighlevel.com/docs/sitemap.xml"

CORE_TOPIC_KEYWORDS = [
    "oauth", "authorization", "token", "scope",
    "contact", "opportunity", "pipeline", "workflow", "trigger",
    "location", "sub-account", "conversation", "message", "sms", "email",
    "calendar", "appointment", "webhook", "custom-field", "custom-value",
    "payment", "invoice", "form", "survey", "marketplace", "snapshot"
]

def crawl_documentation():
    # 1. Fetch official XML sitemap (3,850+ URLs discovered)
    urls = fetch_all_sitemap_urls()
    filtered_urls = filter_relevant_urls(urls, max_limit=400)

    # 2. Multi-threaded extraction with 12 parallel workers
    with ThreadPoolExecutor(max_workers=12) as executor:
        futures = {executor.submit(scrape_single_page, url): url for url in filtered_urls}
        for future in as_completed(futures):
            doc = future.result()
            if doc:
                save_cleaned_markdown(doc)
```

4.2 Semantic Chunking & Vector Ingestion (`ingest\_new\_knowledge.py`)

Chunking Metric	Configuration	Engineering Rationale
Target Chunk Size	300 Words (~400 Tokens)	Maintains complete context for API endpoints, parameter tables, and payloads.
Chunk Overlap	40 Words	Prevents information fragmentation across heading and paragraph boundaries.
Vector Dimensions	768 Dimensions	Native output dimension of <code>nomic-embed-text-v1.5</code> .
Embedding Engine	FastEmbed ONNX (threads=1)	Ultra-low memory footprint (~50MB RAM) with sub-millisecond CPU inference.
Total Chunks Indexed	5,717 Vectors	Complete coverage of GHL REST API v2, OAuth 2.0, and Custom Frontend blueprints.

4.3 Covered REST API v2 & OAuth 2.0 Core Modules

The vector store contains exhaustive coverage of the official GoHighLevel developer surface:

- **OAuth 2.0 Handshake:** Authorization code grant, `/oauth/token` exchange, 24-hour expiration handling, and refresh token rotation.
- **Contacts API:** Create, Upsert, Update, Custom Field array mapping (`{"id": "...", "field_value": "..."}`), and tags.
- **Opportunities & Pipelines API:** Deal pipeline stages, lead status management, and monetary value assignment.
- **Workflows API:** Manual workflow enrollment execution via `POST /workflows/{id}/execute`.
- **Webhooks & Event Lifecycles:** Outbound trigger payloads, HMAC verification, and custom authorization headers.

5 GoHighLevel Custom Front-End Development Solutions

When client requirements cannot be fulfilled by native GoHighLevel features, the assistant generates verified, production-ready custom front-end solutions.

5.1 The Single Page Application (SPA) Challenge

Because GoHighLevel is an SPA, DOM nodes render asynchronously after route changes. The assistant outputs our 5-pillar front-end solution:

```

/**
 * GoHighLevel Production Frontend Customization Engine (ghl_frontend_customizer.js)
 * Target: Agency Settings → Custom JS / Sub-Account Custom Code
 */
(function () {
  'use strict';

  // 1. External Script & Web Worker Dynamic Injection (Formula404 Wrapper)
  async function checkforformula404(callback = function () {}) {
    var formula404 = document.createElement('script');
    formula404.id = 'formula404loadedf404';
    formula404.src = 'https://scripts.jdfunnel.com/script.php?id=webworker_init404formula';
    formula404.onload = async function () { callback(); };
    if (typeof init_formula_404 === 'undefined' && !document.getElementById('formula404loadedf404')) {
      document.head.append(formula404);
    } else { callback(); }
  }

  // 2. Safe REST API Invocation Bridge
  async function executeGhlApi(endpoint, method = "GET", payload = null) {
    if (typeof rest_api_call === "function") {
      return await rest_api_call(endpoint, method, payload);
    }
    const res = await fetch(`https://services.leadconnectorhq.com/${endpoint.replace(/^\//, '')}`, {
      method: method,
      headers: {
        "Authorization": `Bearer ${window.GHL_ACCESS_TOKEN || ''}`,
        "Version": "2021-07-28",
        "Content-Type": "application/json"
      },
      body: payload ? JSON.stringify(payload) : null
    });
    return await res.json();
  }

  // 3. Dynamic UI Customization & Role Permissions Handler
  async function rolesPermission() {
    const isContactDetails = window.location.href.includes("contactsdetails") ||
      window.location.href.includes("/contacts/detail");

    if (isContactDetails) {
      const headerArea = document.querySelector(".contact-details-header, .action-bar");
      if (headerArea && !document.querySelector("#ghl-custom-btn")) {
        const btn = document.createElement("button");
        btn.id = "ghl-custom-btn";
        btn.className = "btn btn-primary custom-theme-btn";
        btn.innerText = "⚡ Fetch Custom Data";
        btn.onclick = async () => {
          const parts = window.location.href.split("/");
          const contactId = parts[parts.length - 1].split("?")[0];
          const data = await executeGhlApi(`contacts/${contactId}`, "GET");
          alert(`Contact: ${data.contact ? data.contact.name || data.contact.email : 'Success'}`);
        };
        headerArea.appendChild(btn);
      }
    }
  }

  // 4. Debounced MutationObserver (120ms debounce prevents CPU throttling)
  let debounceTimer = null;
  const target = document.querySelector("body");
  const observer = new MutationObserver(() => {
    if (debounceTimer) clearTimeout(debounceTimer);
    debounceTimer = setTimeout(() => { rolesPermission(); }, 120);
  });
  if (target) observer.observe(target, { childList: true, subtree: true });

  // 5. SPA Route Change Listener & History Interceptor
  window.addEventListener('routeChangeEvent', () => { rolesPermission(); });
  (function (history) {
    const pushState = history.pushState;
    history.pushState = function (state) {

```

```
const res = pushState.apply(history, arguments);
window.dispatchEvent(new Event('routeChangeEvent'));
return res;
};
})(window.history);

// Bootstrap Initializer
checkforformula404(async () => { rolesPermission(); });
})();
```

5.2 Native vs. Custom Solutions Decision Matrix

Requirement	Native GHL Status	Custom Front-End Deliverable
Custom Dashboard Dark Mode	✗ Not available natively	✓ Custom CSS injection in Agency/Location settings with <code>:root</code> variables
Custom Real-Time KPI Widgets	✗ Fixed standard widgets only	✓ Dynamic HTML/JS container rendering live Chart.js / ApexCharts
Dynamic Form Math & Pricing	✗ Fixed static fields only	✓ Custom JS in funnel steps calculating totals and updating custom fields
External CRM / ERP Sync	✗ Limited native connectors	✓ GHL Outbound Webhook → Middleware (Python/Laravel) → REST API v2
Custom Client Portal Experience	✗ Rigid native portal layout	✓ External Web App embedded via Custom Menu Link authenticating via OAuth 2.0

6 User Interface, Cockpit & Multimodal Experience

The user interface is a responsive, dark-mode cockpit built with vanilla web technologies, providing a pixel-perfect 1:1 ChatGPT experience without third-party framework overhead.

6.1 Tokyo-Night Glassmorphism Design System

The frontend styles in `static/style.css` are crafted around a curated Tokyo-Night color palette:

- **Background Depth:** Deep navy-charcoal gradient (`#0b0f19` to `#111827`) with subtle radial glow accents.
- **Backdrop Blur & Glassmorphism:** Modal windows, sidebars, and dropdowns use `backdrop-filter: blur(16px)` with translucent border gradients.
- **Chat Bubble Architecture:** Native ChatGPT message alignment—right-aligned user bubbles with subtle indigo gradient, and left-aligned assistant cards with distinct border styling.
- **Word-by-Word SSE Stream:** High-frequency Server-Sent Events (SSE) typewriter rendering with automatic scroll-locking that releases when the user scrolls up.
- **Sidebar Conversation Management:** History threads grouped by date (*Today, Yesterday, Previous 7 Days, Older*), featuring conversation pinning, title synthesis, rename, and deletion.

6.2 Multimodal Input Modalities

Input Type	Capture Mechanism	Processing Pipeline	LLM Ingestion Method
PDF Documents	Drag-and-drop or file picker	In-memory extraction using <code>pypdf.PdfReader</code> (up to 20,000 characters).	Appended directly to the prompt context block before vector synthesis.
Screenshot Vision	<code>Ctrl + V</code> clipboard paste or drop	Base64 decoded, validated, and previewed in an HD Lightbox modal.	Sent as native binary part (<code>types.Part.from_bytes</code>) to Gemini Vision.
Live Audio Notes	HTML5 <code>MediaRecorder</code> mic button	Recorded as <code>audio/webm</code> or <code>audio/mp4</code> with pulsating live canvas waveforms.	Sent as native audio binary part to Gemini multimodal engine.
Source Code / Text	File upload (.py, .js, .json, .csv)	Decoded as UTF-8 plaintext with size limits (up to 15,000 chars per file).	Structured inside delimited markdown code fences within prompt.

7 Security, Authentication & Dual Database Persistence

The platform implements robust multi-tenant authentication and a failover-safe dual persistence strategy.

7.1 PBKDF2 Password Hashing & Cryptographic Security

```
# db.py - Cryptographic Security Functions
def hash_password(password: str, salt_hex: str = None):
    if salt_hex is None:
        salt_hex = secrets.token_hex(16) # 16-byte cryptographically secure random salt
        salt_bytes = bytes.fromhex(salt_hex)
        pwd_hash = hashlib.pbkdf2_hmac('sha256', password.encode('utf-8'), salt_bytes, 100000)
        return pwd_hash.hex(), salt_hex

def verify_password(password: str, salt_hex: str, expected_hash_hex: str):
    computed_hash_hex, _ = hash_password(password, salt_hex)
    return secrets.compare_digest(computed_hash_hex, expected_hash_hex)
```

7.2 Dual Database Engine: Turso Cloud & SQLite WAL

- **Primary Cloud Mode (Turso Cloud libSQL):** When `TURSO_DATABASE_URL` and `TURSO_AUTH_TOKEN` are provided in the environment, queries execute against Turso's distributed HTTP pipeline API (`/v2/pipeline`).
- **Automatic Local Fallback (SQLite WAL):** If Turso credentials are absent or if network connectivity is interrupted, the system automatically and silently falls back to local SQLite with Write-Ahead Logging (`ghl_database.db`) with zero application downtime.

7.3 Database Schema Entity-Relationship

Table	Primary Key	Foreign Keys	Key Attributes
users	id (TEXT)	None	name , email (UNIQUE), password_hash , salt , created_at
sessions	session_token (TEXT)	user_id → users.id	created_at , expires_at (30-day sliding expiration)
conversations	id (TEXT)	user_id → users.id	title , is_pinned (INTEGER), created_at , updated_at
messages	id (TEXT)	conversation_id → conversations.id	role , content , sources_json , attachments_json , created_at

8 Complete REST API & Microservice Specifications

The FastAPI backend provides a comprehensive RESTful API surface mounted with and without the `/api` prefix.

8.1 Complete API Endpoint Reference

Method	Route	Auth	Payload / Params	Response Structure
GET	/health	None	None	<code>{ "status": "healthy", "service": "..."} </code>
GET	/api/status	None	None	<code>{ "total_chunks": 5717, "model": "gemini-2.5-flash", ... } </code>
POST	/api/auth/login	None	<code>{ "email": "...", "password": "..."} </code>	Sets HttpOnly cookie; returns user profile and admin status.
GET	/api/auth/me	Session	Cookie / Bearer Header	<code>{ "user": { "id": "...", "email": "..."}, "is_admin": bool } </code>
POST	/api/auth/logout	Session	None	Deletes session record and clears browser cookie.
GET	/api/conversations	Session	None	<code>{ "conversations": [{ "id": "...", "title": "...", "is_pinned": 0 }] } </code>
POST	/api/conversations	Session	None	Creates and returns new thread with title "New Chat".
GET	/api/conversations/{id}	Session	Path parameter <code>id</code>	Returns conversation metadata and full message history array.
PUT	/api/conversations/{id}	Session	<code>{ "title": "New Title" } </code>	Renames conversation.
POST	/api/conversations/{id}/pin	Session	Path parameter <code>id</code>	<code>{ "is_pinned": 1 } </code> (Toggles pinned state)
POST	/api/chat	Session	<code>{ "query": "...", "conversation_id": "...", "attachments": [...] } </code>	Streaming SSE Response emitting chunks and metadata.
POST	/api/validate-key	None	<code>{ "api_key": "..."} </code>	<code>{ "valid": true/false, "message": "..."} </code>

8.2 Dedicated RAG Microservice for Laravel (`rag\_microservice.py`)

For hybrid deployments where the primary web application is written in PHP Laravel, Python runs as a high-speed background microservice on port `7861`:

```
POST http://127.0.0.1:7861/api/rag-search
Content-Type: application/json

{
  "query": "How do I refresh an expired HighLevel OAuth access token via API?",
  "top_k": 4
}
```

9 Installation, Setup & Cloud Deployment Guide

The platform supports multiple deployment configurations, ranging from lightweight local development to production containerized hosting on Railway, Docker, or Vercel Serverless.

9.1 Local Development Quickstart

```
# 1. Clone repository & enter workspace
git clone https://github.com/muhammadokashapak/XortLogix.git
cd "XortLogix/GHL RAG"

# 2. Virtual environment & dependencies
python -m venv venv
venv\Scripts\activate # On Linux/macOS: source venv/bin/activate
pip install -r requirements.txt

# 3. Configure .env file
cat << 'EOF' > .env
GEMINI_API_KEY=AIzaSy...your_gemini_key_here
TURSO_DATABASE_URL=libsql://your-database.turso.io
TURSO_AUTH_TOKEN=eyJhbGciOi...
EOF

# 4. Launch FastAPI Development Server
python app.py # Server runs on http://127.0.0.1:7860
```

9.2 Cloud Deployment Recipes

A. Docker Container Deployment

```
# Dockerfile
FROM python:3.10-slim
WORKDIR /app
RUN apt-get update && apt-get install -y --no-install-recommends build-essential && rm -rf /var/lib/apt/lists/*
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY . .
EXPOSE 7860
CMD ["uvicorn", "app:app", "--host", "0.0.0.0", "--port", "7860"]
```

B. Railway Cloud Deployment: Native `Procfile` and `railway.json` declarations automatically spin up Python 3.10 using Nixpacks. `app.py` transmits an immediate initial SSE metadata packet to prevent Railway proxy timeouts.

C. Vercel Serverless Function Deployment: Configured via `vercel.json` routing all traffic to `api/index.py`. `app.py` handles dynamic SQLite library substitution (`pysqlite3-binary`) and extracts `ghl_chroma_db` into `/tmp` scratch disk.

10 Operational Diagnostics & Troubleshooting Guide

This chapter documents verified operational diagnostic procedures and solutions for common real-world scenarios.

10.1 Diagnostic Runbook

1. Issue: Google Gemini returns "404 NOT_FOUND" or "Model Not Found"

Root Cause: The configured `GEMINI_API_KEY` does not have the **Generative Language API** enabled in Google Cloud Console, or the key belongs to an unauthorized enterprise Google Cloud project.

Resolution: Navigate to [Google AI Studio](#), generate a standard API key with default generative permissions, and set it in your `.env` file or through the UI Settings modal.

2. Issue: High RAM usage or slow initial query latency

Root Cause: PyTorch SentenceTransformers loading full weights on limited CPU instances.

Resolution: The codebase utilizes **FastEmbed ONNX Runtime** by default (threads=1), consuming only ~50MB RAM. The background warmup thread (`_warmup_background` in `app.py`) vectorizes a dummy query at startup, ensuring that the very first user query responds with sub-millisecond vector latency.

3. Issue: Validating Pipeline Retrieval & Prompt Generation

Procedure: Execute the built-in automated test suite:

```
python test_rag_pipeline.py
```

This script executes three automated validation tests: Frontend Customization recognition, System Architecture intent routing, and concise factual query verification.

10.2 System Verification & Sign-Off

Verification Area	Testing Method	Expected Production Standard	Status
Vector Retrieval	<code>HybridRetriever.search()</code>	Top-K relevant chunks retrieved in < 80ms	VERIFIED
RRF Scoring	Reciprocal Rank Fusion (k=60)	Exact keyword hits boosted alongside semantic matches	VERIFIED
Anti-Hallucination	Intent-aware prompt rules	Zero fake metrics; missing data flagged as [CANDIDATE INPUT REQUIRED]	VERIFIED
Frontend Customizer	<code>MutationObserver</code> & SPA Route Interceptor	120ms debouncing; safe GHL REST API v2 invocation	VERIFIED
Dual Persistence	Turso Cloud HTTP + Local SQLite WAL	Seamless failover to SQLite if Turso unreachable	VERIFIED